

Question-1

$$u = \begin{bmatrix} 1 \\ 0 \\ 1 \\ 1 \end{bmatrix}, \quad v = \begin{bmatrix} 0 \\ 2 \\ 1 \\ 1 \end{bmatrix} \quad \& \quad w = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 3 \end{bmatrix}$$

(i) Determine 2 vectors most similar to each other:

(a) l_2 norm

$$\text{dist}(x, y) = \|x - y\|_2 = \|y - x\|_2 = \sqrt{\sum_{i=1}^n (x_i - y_i)^2}$$

$$\text{dist}(u, v) = \sqrt{(1-0)^2 + (0-2)^2 + (1-1)^2 + (1-1)^2} = \sqrt{5} \approx 2.236$$

$$\text{dist}(u, w) = \sqrt{(1-1)^2 + (0-1)^2 + (1-0)^2 + (1-3)^2} = \sqrt{6} \approx 2.449$$

$$\text{dist}(v, w) = \sqrt{(0-1)^2 + (2-1)^2 + (1-0)^2 + (1-3)^2} = \sqrt{7} \approx 2.645$$

Most similar vectors are $\vec{u}$ & $\vec{v}$ as per l_2 norm.

b) l_1 norm

$$\text{dist}(x, y) = \|x - y\|_1 = \|y - x\|_1 = \sum_{i=1}^n |x_i - y_i|$$

$$\text{dist}(u, v) = |1-0| + |0-2| + |1-1| + |1-1| = 3$$

$$\text{dist}(u, w) = |1-1| + |0-1| + |1-0| + |1-3| = 4$$

$$\text{dist}(v, w) = |0-1| + |2-1| + |1-0| + |1-3| = 5$$

Most similar vectors are $\vec{u}$ & $\vec{v}$ as per l_1 norm.

(ii) Determine 2 vectors most similar based on cosine similarity

$$\cos(\theta) = \frac{x \cdot y}{\|x\| \|y\|}$$

$$\cos(\theta)_{uv} = \frac{1 \times 0 + 0 \times 2 + 1 \times 1 + 1 \times 1}{\sqrt{1^2 + 0^2 + 1^2 + 1^2} \sqrt{0^2 + 2^2 + 1^2 + 1^2}} = \frac{2}{\sqrt{3} \sqrt{6}} \approx 0.471$$

$$\cos(\theta)_{uw} = \frac{1 \times 1 + 0 \times 1 + 1 \times 0 + 1 \times 3}{\sqrt{1^2 + 0^2 + 1^2 + 1^2} \sqrt{0^2 + 2^2 + 1^2 + 1^2}} = \frac{4}{\sqrt{3} \sqrt{6}} \approx 0.696$$

$$\cos(\theta)_{vw} = \frac{0 \times 1 + 2 \times 1 + 1 \times 0 + 1 \times 3}{\sqrt{0^2 + 2^2 + 1^2 + 1^2} \sqrt{0^2 + 2^2 + 1^2 + 1^2}} = \frac{5}{\sqrt{6} \sqrt{6}} \approx 0.615$$

Most similar vectors are $\vec{u}$ & $\vec{w}$ as per cosine similarity.

(iii) Reason behind the difference in result b/w (i) & (ii)

As l_1 and l_2 norms are applied over vector magnitude of differences in vector component, that's why got $\vec{v}$ & $\vec{w}$ most similar.

But, cosine similarity focus on angle b/w vectors, that's why got $\vec{v}$ & $\vec{u}$ most similar.

(iv) Resolve difference using unit vector

$$v_1 = \frac{1}{\|v\|} \vec{v} = \begin{bmatrix} 1/\sqrt{5} \\ 0 \\ 1/\sqrt{5} \\ 1/\sqrt{5} \end{bmatrix}$$

$$v_2 = \frac{1}{\|v\|} \vec{v} = \begin{bmatrix} 0 \\ 2/\sqrt{6} \\ 1/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}, \quad w_1 = \frac{1}{\|w\|} \vec{w} = \begin{bmatrix} 1/\sqrt{11} \\ 1/\sqrt{11} \\ 0 \\ 3/\sqrt{11} \end{bmatrix}$$

Calculating l_2 norm

$$\text{dist}(v_1, v_2) = 1.028$$

$$\text{dist}(v_1, w_1) = 0.779$$

$$\text{dist}(v_2, w_1) = 0.876$$

Calculating l_1 norm

$$\text{dist}(v_1, v_2) = 1.732$$

$$\text{dist}(v_1, w_1) = 1.481$$

$$\text{dist}(v_2, w_1) = 1.721$$

So, as per the above observation, $\vec{v}$ & $\vec{w}$ are most similar as per l_1 & l_2 norms and cosine similarity.

So this has been resolved via unit vector.

After normalizing the vectors to unit length, magnitude based distance & direction-based similarity are similar.

Question-2

$$A = \begin{bmatrix} -1 & 0 & a \\ b & 4 & c \\ d & 0 & 0 \end{bmatrix}$$

(i) Rank of A based on a, b, c & d

Using row echelon form:

$$\begin{bmatrix} -1 & 0 & a \\ b & 4 & c \\ d & 0 & 0 \end{bmatrix}$$

$$R_2 \rightarrow R_2 + bR_1$$

$$R_3 \rightarrow R_3 + dR_1$$

$$\begin{bmatrix} -1 & 0 & a \\ 0 & 4 & c+4b \\ 0 & 0 & ad \end{bmatrix}$$

if a or d is 0, then Rank is 2

else, Rank is 3

So, rank A depends on values of a & d.

(ii) Number of distinct eigen values

$$A - \lambda I = 0$$

using original A value,

$$A - \lambda I = \begin{bmatrix} -1 & 0 & a \\ b & 4 & c \\ d & 0 & 0 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} -1-\lambda & 0 & a \\ b & 4-\lambda & c \\ d & 0 & -\lambda \end{bmatrix}$$

$$A - \lambda I = \begin{bmatrix} -1-\lambda & 0 & a \\ b & 4-\lambda & c \\ d & 0 & -\lambda \end{bmatrix} \neq \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\det(A - \lambda I) = (1+\lambda)(4-\lambda)\lambda + a(4-\lambda)(-d)$$

$$= (4-\lambda)(\lambda^2 + \lambda - 4ad)$$

$$\text{So, } \lambda = 4 \quad \text{or} \quad \lambda = \frac{-1 \pm \sqrt{1+4ad}}{2}$$

if $ad=0$, then λ will have values as } result
4, 0 & -1

else, λ will have values as $4, \frac{1+\sqrt{1+4ad}}{2}, \frac{1-\sqrt{1+4ad}}{2}$ result

Using REF result obtained from (i)

$$A - \lambda I$$

$$\begin{bmatrix} -1 & 0 & a \\ 0 & 4 & c+ab \\ 0 & 0 & ad \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (ii)$$

$$\begin{bmatrix} -1-\lambda & 0 & a \\ 0 & 4-\lambda & c+ab \\ 0 & 0 & ad-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = -(1+\lambda)(4-\lambda)(ad-\lambda)$$

$$\text{so, } \lambda = -1 \text{ or } \lambda = 4 \text{ or } \lambda = ad$$

if $ad=0$, then λ is same as calculated above.

$$4, 0 \text{ and } -1$$

Question-3

$$A = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

$$A^T = \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix}$$

$$(i) Ax = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, x \in \mathbb{R}^3$$

$$\begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} -x_1 + x_2 \\ -x_2 + x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\text{so, } -x_1 + x_2 = 1 \Rightarrow x_1 = x_2 - 1$$

$$\text{So, } -x_2 + x_3 = 1 \Rightarrow x_3 = 1 + x_2$$

from above derivation

$$x = \begin{bmatrix} x_2 - 1 \\ x_2 \\ x_2 + 1 \end{bmatrix} = x_2 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \text{ for any } x_2 \in \mathbb{R}$$

(ii) Rank(A)

Using REF

$$\begin{aligned} \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix} & \begin{aligned} R_1 &\rightarrow -R_1 \\ R_2 &\rightarrow -R_2 \\ R_1 &\rightarrow R_1 + R_2 \end{aligned} \\ \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \end{bmatrix} & \end{aligned}$$

From above row reducing A, rank(A) = 2.

✓ Basis for column space C(A)

As 1st 2 columns of A are linearly independent, so

$$C(A) = \left\{ \begin{bmatrix} -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

✓ Basis for row space C(A^T)

$$C(A^T) = \left\{ \begin{bmatrix} -1 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 & 1 \end{bmatrix} \right\} = \left\{ \begin{bmatrix} 1 & 0 & -1 \end{bmatrix}, \begin{bmatrix} 0 & 1 & -1 \end{bmatrix} \right\}$$

✓ Basis for null space N(A)

$$Ax = 0$$

$$\begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$\text{So, } x_1 - x_2 = 0, \quad -x_2 + x_3 = 0$$

$$x_2 = x_1$$

$$x_2 = x_3$$

$$\text{means, } x_1 = x_2 = x_3$$

$$N(A) = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$$

✓ Basis for left null space $N(A^T)$

$$A^T y = 0$$

$$\begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} y = 0$$

~~Simplify~~ Simplifying matrix

$$\left[\begin{array}{cc|c} -1 & 0 & 0 \\ 1 & -1 & 0 \\ 0 & 1 & 0 \end{array} \right]$$

from above, we derive

$$y_1 = 0 \text{ \& } y_2 = 0$$

So $N(A^T)$ is trivial & spanned by $\left\{ \begin{bmatrix} 0 \\ 0 \end{bmatrix} \right\}$ or $\{0\}$

(iii) Construct $B = AA^T$, & find eigenvalues & eigenvectors of B

$$B = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$$

$$\det(B - \lambda I) = 0$$

$$\det\left(\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}\right) = 0$$

$$\det\left(\begin{bmatrix} 2-\lambda & -1 \\ -1 & 2-\lambda \end{bmatrix}\right) = 0$$

$$(2-\lambda)^2 - 1 = 0$$

$$\lambda^2 - 4\lambda + 3 = 0$$

$$(\lambda-3)(\lambda-1) = 0$$

✓ So, eigenvalues are $\lambda_1 = 3$ & $\lambda_2 = 1$

✓ Eoⁿ $\lambda_1 = 3$

$$(B - 3I)x = 0$$

$$\begin{bmatrix} -1 & -1 \\ -1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Simplifying, $x_1 = -x_2$, if $x_1 = 1$, then $x_2 = -1$
 So, 1 eigenvector is

$$✓ v_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

✓ Eoⁿ $\lambda_2 = 1$

$$(B - I)x = 0$$

$$\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Simplifying, $x_1 = x_2$, if $x_1 = 1$, then $x_2 = 1$

So, eigenvector is

$$✓ v_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Construct matrix D

& $\sigma_1 = \sqrt{\lambda_1}$, $\lambda_1 \geq \lambda_2$

$$\text{as, } \lambda_1 = 3 \text{ \& } \lambda_2 = 1$$

$$\text{So, } \sigma_1 = \sqrt{3} \text{ \& } \sigma_2 = 1$$

$$D = \begin{bmatrix} \sigma_1 & 0 & 0 \\ 0 & \sigma_2 & 0 \end{bmatrix}$$

$$✓ D = \begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \end{bmatrix}$$

(iv) Using Gram-Schmidt process

→ Normalise v_1

$$✓ \hat{u}_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

→ Orthogonalize v_2 w.r.t u_1

$$v_2' = v_2 - \text{proj}_{u_1} v_2$$

$$\text{proj}_{u_1} u_2 = \left(\frac{u_2 \cdot u_1}{u_1 \cdot u_1} \right) u_1$$

$$u_2 \cdot u_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} = 0$$

$\therefore u_2 \cdot u_1 = 0$, u_2 is orthogonal to u_1 , so, $u_2' = u_2$.

— Normalize u_2

$$\checkmark \hat{u}_2 = \frac{u_2}{\|u_2\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\checkmark U = \begin{bmatrix} u_1 & u_2 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

— Checking U is an orthogonal matrix

$$\begin{aligned} U^T U &= \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{2} + \frac{1}{2} & \frac{1}{2} - \frac{1}{2} \\ \frac{1}{2} - \frac{1}{2} & \frac{1}{2} + \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \end{aligned}$$

Thus, U is an orthogonal matrix

(v) & (vi) Construct $G = A^T A$

$$\checkmark G = \begin{bmatrix} -1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

Find eigenvalue & eigenvectors of G

$$\det(G - \lambda I) = 0$$

$$\det \left(\begin{bmatrix} 1-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 1-\lambda \end{bmatrix} - \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right) = 0$$

$$\det \left(\begin{bmatrix} 1-\lambda & -1 & 0 \\ -1 & 2-\lambda & -1 \\ 0 & -1 & 1-\lambda \end{bmatrix} \right) = 0$$

$$(1-\lambda)[(2-\lambda)(1-\lambda)-1]-1(1-\lambda)=0$$

$$(1-\lambda)(2-2\lambda-\lambda+\lambda^2+2)=0$$

$$(1-\lambda)(\lambda^2-3\lambda)=0$$

$$\lambda(1-\lambda)(\lambda-3)=0$$

$$\checkmark \text{ So, } \lambda_1 = 3, \lambda_2 = 1, \lambda_3 = 0$$

For $\lambda_1 = 3$

$$(A - 3I)x = 0$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ -1 & 2 & -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array} \right] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -2 & -1 & 0 \\ -1 & -1 & -1 \\ 0 & -1 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{So, } -2x_1 - x_2 = 0 \Rightarrow x_2 = -2x_1$$

$$-x_1 - x_2 - x_3 = 0$$

$$-x_2 - 2x_3 = 0 \Rightarrow x_2 = -2x_3$$

So, eigenvector of $\lambda_1 = 3$ is

$$\checkmark v_1 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

For $\lambda_2 = 1$

$$(A - I)x = 0$$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 0 & 1 & 0 & 0 \\ -1 & 2 & -1 & 0 & 1 & 0 \\ 0 & -1 & 1 & 0 & 0 & 1 \end{array} \right] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 0 & -1 & 0 \\ -1 & 1 & -1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{So, } -x_2 = 0 \Rightarrow x_2 = 0$$

$$-x_1 + x_2 - x_3 = 0$$

$$-x_1 - x_3 = 0$$

So, eigenvector of $\lambda_2 = 1$ is

$$v_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

For $\lambda_3 = 0$

$$(A - 0I)x = 0$$

$$\begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$\text{So, } x_1 - x_2 = 0$$

$$\Rightarrow x_2 = x_1$$

$$-x_1 + 2x_2 - x_3 = 0$$

$$-x_2 + x_3 = 0$$

$$\Rightarrow x_2 = x_3$$

So, eigenvector of $\lambda_3 = 0$ is

$$v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Orthogonalizing eigenvectors

Normalize v_1 :

$$v_1 = \frac{v_1}{\|v_1\|} = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix}$$

Orthogonalize v_2 w.r.t v_1

$$v_2' = v_2 - \text{proj}_{v_1} v_2$$

$$\text{proj}_{v_1} v_2 = \left(\frac{v_2 \cdot v_1}{v_1 \cdot v_1} \right) v_1$$

$$v_2 \cdot v_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \cdot \begin{bmatrix} 1/\sqrt{6} \\ -2/\sqrt{6} \\ 1/\sqrt{6} \end{bmatrix} = 0$$

$\therefore v_2 \cdot v_1 = 0$, v_2 is orthogonal to v_1 , So, $v_2' = v_2$

Normalize v_2

$$v_2 = \frac{v_2}{\|v_2\|} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ 0 \\ -\frac{1}{\sqrt{2}} \end{bmatrix}$$

Orthogonalize v_3 w.r.t v_1

$$v_3' = v_3 - \text{proj}_{v_1} v_3$$

$$\text{proj}_{v_1} v_3 = \left(\frac{v_3 \cdot v_1}{v_1 \cdot v_1} \right) v_1$$

$$v_3 \cdot v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \cdot \begin{bmatrix} \frac{1}{\sqrt{6}} \\ -\frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{6}} \end{bmatrix} = 0$$

$\therefore v_3 \cdot v_1 = 0$, v_3 is orthogonal to v_1 , so, $v_3' = v_3$

Normalize v_3

$$v_3 = \frac{v_3}{\|v_3\|} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{3}} \end{bmatrix}$$

$$V = [v_1 \ v_2 \ v_3]$$

$$V = \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ -\frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{bmatrix}$$

Checking V is orthogonal matrix

$$V^T V = I$$

$$\begin{aligned} V^T V &= \begin{bmatrix} \frac{1}{\sqrt{6}} & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & 0 & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \\ -\frac{2}{\sqrt{6}} & 0 & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{3}} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{6} + \frac{4}{6} + \frac{1}{6} & \frac{1}{\sqrt{12}} + 0 - \frac{1}{\sqrt{12}} & \frac{1}{\sqrt{18}} - \frac{2}{\sqrt{18}} + \frac{1}{\sqrt{18}} \\ \frac{1}{\sqrt{12}} + 0 - \frac{1}{\sqrt{12}} & \frac{1}{2} + 0 + \frac{1}{2} & \frac{1}{\sqrt{6}} + 0 - \frac{1}{\sqrt{6}} \\ \frac{1}{\sqrt{18}} - \frac{2}{\sqrt{18}} + \frac{1}{\sqrt{18}} & \frac{1}{\sqrt{6}} + 0 - \frac{1}{\sqrt{6}} & \frac{1}{3} + \frac{1}{3} + \frac{1}{3} \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \end{aligned}$$

Thus, V is orthogonal matrix.

(vii) $w_i = \frac{1}{\sigma_i} A^T u_i, \quad i=1,2$

$$u_1 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$$u_2 = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$$

$$\sigma_1 = \sqrt{3}$$

$$\sigma_2 = 1$$

$$w_1 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} -1/\sqrt{2} \\ 2/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix}$$

$$w_1 = \begin{bmatrix} -1/\sqrt{6} \\ 2/\sqrt{6} \\ -1/\sqrt{6} \end{bmatrix}$$

$$w_2 = \frac{1}{1} \begin{bmatrix} -1 & 0 \\ 1 & -1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix}$$

Find w_3 , as $w_3 \perp w_1$, $w_3 \perp w_2$ & $\|w_3\| = 1$

So, $w_3 = \frac{w_1 \times w_2}{\|w_1 \times w_2\|}$

$$w_1 \times w_2 = \begin{vmatrix} i & j & k \\ -1/\sqrt{6} & 2/\sqrt{6} & -1/\sqrt{6} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{vmatrix}$$

$$= \frac{2i}{\sqrt{12}} + \frac{2j}{\sqrt{12}} + \frac{2k}{\sqrt{12}}$$

$$= \frac{1}{2\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

magnitude of $w_1 \times w_2$

$$\|w_1 \times w_2\| = \sqrt{\left(\frac{1}{2\sqrt{3}}\right)^2 + \left(\frac{1}{2\sqrt{3}}\right)^2 + \left(\frac{1}{2\sqrt{3}}\right)^2} = \frac{\sqrt{3}}{2\sqrt{3}} = \frac{1}{2}$$

$$w_3 = \frac{1/2\sqrt{2}}{\sqrt{3}/2\sqrt{2}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

$$w_3 = \frac{1/2\sqrt{3}}{1/2} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

Let's compare $\{w_1, w_2, w_3\}$ with $\{v_1, v_2, v_3\}$ (iv)

$$w_1 = -v_1$$

$$w_2 = -v_2$$

$$w_3 = v_3$$

- Vectors in $\{w_1, w_2, w_3\}$ are scaled versions of those in $\{v_1, v_2, v_3\}$, but with opposite signs for w_1 & w_2 .
- w_3 & v_3 are identical, indicating they represent the same direction in R^3 .
- Both sets are orthonormal sets.
- Both sets form an orthonormal basis for R^3 .
- Vectors in each set have similar components but may differ by sign.

(viii) Compute $U D V^T$

$$U = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}, \quad D = \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$$V = \begin{bmatrix} 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ -2/\sqrt{6} & 0 & 1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \end{bmatrix}, \quad V^T = \begin{bmatrix} 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix}$$

$$\begin{aligned} D \cdot V^T &= \begin{bmatrix} \sqrt{3} & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{6} & -2/\sqrt{6} & 1/\sqrt{6} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \\ 1/\sqrt{3} & 1/\sqrt{3} & 1/\sqrt{3} \end{bmatrix} \\ &= \begin{bmatrix} \frac{\sqrt{3}}{\sqrt{6}} & -\frac{2\sqrt{3}}{\sqrt{6}} & \frac{\sqrt{3}}{\sqrt{6}} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -2/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \end{bmatrix} \end{aligned}$$

$$U \cdot (D V^T) = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} & -2/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & 0 & -1/\sqrt{2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2} + \frac{1}{2} & -\frac{2}{2} & \frac{1}{2} - \frac{1}{2} \\ -\frac{1}{2} + \frac{1}{2} & \frac{2}{2} & -\frac{1}{2} - \frac{1}{2} \end{bmatrix} = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \end{bmatrix}$$

(ix) Columns of U & (CA^T)

- There is no direct 1-1 correspondence in terms of components, but both sets ~~are~~ represent orthogonal bases in their spaces.

Columns of V & $C(A)$

- Columns of V are orthonormal vectors related to right singular vectors of A , which span the same space as columns of A .

Columns of V & $N(A)$

- Proportional to each other, indicating that 3rd column of V represents the null space of A .

$$V(3^{\text{rd}} \text{ column}) = \begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix} \quad N(A) = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

Columns of U & $N(A^T)$

- Not directly related by scalar multiplication, but U captures left singular vectors which span orthogonal complement to row space of A .

Thus, the SVD of A through U, D & V provides orthonormal bases for the 4 fundamental subspaces of A .